

C# Weekly II COD Practice Assignment: Library Book Management System

Problem Statement

Create a **C# Console Application** to manage books in a small library. The librarian should be able to **add books, search for a book, display all books, and delete a book** with proper validations and exception handling.

Write the complete solution inside the **Program.cs** file.

Requirements

Class: Library

Create a class called Library.

Properties

List<string> bookList

Used to store book names in the library.

const int TotalBooks = 5;

Used to set the maximum number of books allowed in the library.

Methods

1. AddBook(string bookName)

This method should add a book to the bookList.

Validations

1. Book name should not be null, empty, or whitespace.

If invalid, throw:

ArgumentException

Message:

Book name cannot be null, empty or whitespace

2. Total books should not be more than TotalBooks.

If the limit is exceeded, throw:

InvalidOperationException

Message:

Cannot add more than 5 books

2. ShowAllBooks()

Display all books available in the library.

If no books are available, display:

No books available in the library

3. FindBook(string bookName)

Search for a book from the library.

If found, display:

Book Found: <bookName>

Otherwise, display:

Book Not Found

4. DeleteBook(string bookName)

Delete the given book from the library.

If book exists, delete and display:

Book deleted successfully

Otherwise, display:

Book not found, cannot delete

Program Class Requirements

In the Main method:

1. Prompt the user to enter a comma-separated list of book names.
2. Add each book to the library using AddBook().
3. Handle all exceptions using try-catch.
4. Prompt the user to enter a book name to search.
5. Call FindBook().
6. Prompt the user to enter a book name to delete.
7. Call DeleteBook().
8. Finally display all books using ShowAllBooks().

Input Format

A single line containing comma-separated book names
A second line containing the book name to search
A third line containing the book name to delete

Sample Input

CSharp Basics,SQL Server,ASP.NET Core,Angular Guide
SQL Server
Angular Guide

Expected Output

Book added successfully: CSharp Basics
Book added successfully: SQL Server
Book added successfully: ASP.NET Core
Book added successfully: Angular Guide

Book Found: SQL Server

Book deleted successfully

All Books:
CSharp Basics
SQL Server
ASP.NET Core

Constraints

Use List<string> Property
Use const int TotalBooks = 5
Use ArgumentException for invalid book name
Use InvalidOperationException for exceeding book limit
Use try-catch for exception handling
Use menu is not required
Write all code inside Program.cs